

ELEOS Adventure

An Accessible Off-Road Mobility Platform Designed for Rugged Terrain

Hero Section

A Low-Cost Off-Road Wheelchair Designed to Restore Mobility, Independence, and Outdoor Accessibility

The ELEOS Adventure project is a senior capstone engineering project developed in partnership with ELEOS Solutions, a nonprofit organization focused on designing affordable mobility systems for underserved communities around the world. The project centered around a major accessibility challenge: traditional wheelchairs perform poorly on rough terrain, while existing off-road wheelchair systems are often prohibitively expensive, difficult to manufacture, or inaccessible to users in developing regions.

The team set out to design a wheelchair platform capable of handling rough terrain while remaining:

- Affordable
- Manufacturable using commercially available components
- Easy to reproduce
- Practical for real-world users

The final prototype successfully integrated:

- Lever-driven propulsion
- Bike tire wheel systems
- Pushrim operation
- Disc brakes
- Quick-release wheel systems
- Modular terrain-focused architecture

The resulting system dramatically improved mobility over difficult terrain while remaining significantly lower cost than many existing commercial alternatives.

What Is ELEOS Solutions?

ELEOS Solutions is a nonprofit engineering organization focused on designing practical mobility systems for underserved populations. Their projects often target low- and middle-income countries (LMICs), where traditional mobility systems fail due to:

- Poor infrastructure
- Rough terrain
- Cost barriers
- Limited repair resources
- Limited access to specialized manufacturing

The organization focuses heavily on:

- Accessibility
- Affordability
- Manufacturability
- Long-term serviceability
- Human-centered design

Rather than creating highly specialized systems requiring expensive proprietary components, ELEOS projects prioritize:

- Off-the-shelf parts
- Simplified fabrication
- Local manufacturability
- Repairability

The ELEOS Adventure project directly reflects these goals.

The Problem

Traditional wheelchairs are optimized primarily for smooth indoor surfaces such as:

- Sidewalks
- Pavement
- Indoor flooring
- Ramps

However, many users around the world regularly encounter:

- Dirt roads
- Gravel paths
- Mud
- Sand

- Potholes
- Uneven terrain
- Hills and inclines

For users in these environments, traditional wheelchairs become:

- Extremely difficult to propel
- Physically exhausting
- Unsafe
- Limiting to independence

Existing off-road mobility systems partially solve this issue, but many commercially available systems have major drawbacks.

Existing Solutions Analysis

The team researched existing off-road mobility systems before beginning the design process.

GRIT Freedom Chair

One of the primary benchmark systems analyzed was the GRIT Freedom Chair.

Advantages

- Lever-driven propulsion
- Large front caster wheel
- Strong off-road capability
- Effective terrain handling

Drawbacks

- Expensive
- Uses complex custom-machined parts
- Difficult to reproduce affordably

While the GRIT Freedom Chair demonstrated excellent terrain capability, its cost and manufacturing complexity conflicted with the accessibility goals of the ELEOS project.

Frankie Wheelchair Modification System

The team also analyzed the Frankie system.

Advantages

- Lower cost
- Modifies existing wheelchairs
- Simpler architecture

Drawbacks

- Limited off-road capability
- Marginal terrain improvement without trike modification
- Less stable on difficult terrain

The Frankie system demonstrated that simpler, lower-cost systems were possible, but its terrain performance limitations highlighted the need for a more robust solution.

Project Goals

The project established several key client requirements and engineering goals.

Client Specifications

The final system needed to:

1. Produce a functioning prototype
2. Use commercially available materials
3. Maintain material costs below \$800
4. Be easy to reproduce

These constraints heavily influenced every engineering decision throughout the project.

Design Philosophy

The ELEOS Adventure project was built around several major engineering principles.

1. Accessibility Through Affordability

A major objective was reducing the cost barrier associated with off-road mobility systems.

To accomplish this, the team prioritized:

- Standard bicycle components
- Commercially available materials
- Simplified assemblies

- Reduced custom machining
 - Minimal proprietary parts
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2. Manufacturability

The system needed to be reproducible using relatively simple fabrication methods.

The project intentionally minimized:

- Complex CNC machining
- Specialized tooling
- Difficult assembly procedures

Instead, the design emphasized:

- Welded structures
 - Bolted assemblies
 - Standard bike components
 - Modular construction
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3. Terrain Performance

The wheelchair needed to perform well on:

- Dirt
- Gravel
- Hills
- Uneven terrain
- Potholes

This required careful consideration of:

- Wheel size
 - Weight distribution
 - Stability
 - Braking systems
 - Propulsion efficiency
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4. User Flexibility

The system needed to remain usable in both:

- Off-road environments

- Standard wheelchair operation

This dual-use philosophy became one of the most important design considerations in the project.

Final System Overview

The final ELEOS Adventure prototype is a lever-driven off-road wheelchair platform that combines:

- Large bicycle tires
- Lever-drive propulsion
- Pushrim capability
- Disc brakes
- Quick-release wheel systems
- Detachable drive levers

The design significantly improved terrain capability while preserving traditional wheelchair functionality.

Lever-Driven Propulsion System

One of the defining features of the ELEOS Adventure system is its lever-drive propulsion architecture.

Instead of relying entirely on pushrims, the system uses long drive levers that allow the user to generate significantly greater torque and propulsion force.

Why Lever Drive?

Traditional pushrim wheelchairs become extremely inefficient on rough terrain because:

- Small pushrim diameters reduce leverage
- Traction is reduced
- Users fatigue quickly
- Hill climbing becomes extremely difficult

Lever propulsion improves this by:

- Increasing mechanical advantage
 - Allowing stronger propulsion strokes
 - Improving hill-climbing capability
 - Reducing physical strain
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Detachable Lever System

The drive levers were intentionally designed to be removable.

Key Features

- Quick-detach mechanism
- Allows reverse operation
- Simplifies storage and transport
- Improves user flexibility
- Can be clipped onto the frame when not in use

This modularity allows users to quickly switch between:

- Lever-driven propulsion
- Traditional pushrim operation

The detachable system also improved maintainability and portability.

Wheel System

The wheelchair uses bicycle-style wheels combined with integrated pushrims.

Advantages of Bike Tires

Compared to traditional wheelchair tires, bicycle tires provide:

- Improved traction
- Better obstacle traversal
- Reduced rolling resistance
- Increased propulsion efficiency
- Better off-road performance

The larger wheel diameter significantly improved:

- Pothole traversal
 - Dirt handling
 - Hill climbing
 - Energy efficiency
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Pushrim Integration

Despite the lever-drive focus, maintaining traditional wheelchair functionality remained important.

The integrated pushrims allow:

- Standard wheelchair propulsion
- Indoor usability
- Maneuverability in tight spaces
- User familiarity

This dual-mode operation increased the practicality of the system in real-world environments.

Disc Brake System

The project integrated bicycle-style disc brakes into the wheelchair platform.

Why Disc Brakes?

Traditional wheelchairs often rely on:

- Pushrim friction braking
- User arm strength
- Mechanical resistance

These methods become unreliable on:

- Hills
- Dirt
- Wet terrain
- Loose surfaces

The disc brake system dramatically improved:

- Safety
 - User control
 - Stopping distance
 - Hill climbing confidence
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Hill Climbing Benefits

One particularly important advantage of the brake system was preventing rollback while climbing hills.

Without braking support:

- The wheelchair could roll backward between propulsion strokes
- Users lose efficiency
- Hill climbing becomes dangerous

The braking system allowed users to:

- Maintain position on inclines
 - Climb more efficiently
 - Increase overall safety
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Dog Gear Quick-Release System

One of the more innovative subsystems developed during the project was the quick-release dog gear mechanism.

Purpose

The dog gear system allows:

- Quick wheel removal
- Easier maintenance
- Reduced transport size
- Faster wheel replacement

This feature significantly improves serviceability and portability.

Material Selection

The dog gear was designed using nylon because of its:

- High strength
- Durability
- Suitability for injection molding
- Reduced manufacturing cost

The team also considered future mass-production potential during the design process.

Front Caster System

The front caster wheel plays a major role in terrain traversal.

Traditional wheelchair casters are often small and struggle with:

- Rocks
- Gravel
- Potholes
- Dirt

The larger front caster in the ELEOS Adventure system improved:

- Obstacle traversal
- Stability
- Steering over rough terrain
- Terrain adaptability

Structural Design

The frame architecture was designed around:

- Simplicity
- Strength
- Manufacturability
- Accessibility

The structure needed to support:

- Terrain loads
- User weight
- Propulsion forces
- Braking loads

while remaining practical to fabricate.

Manufacturing Philosophy

A major focus of the project was ensuring the design could be reproduced affordably.

The project intentionally minimized:

- Complex machining
- Custom proprietary components
- Specialized fabrication processes

The final system heavily utilized:

- Standard bike parts
- Welded steel components
- Commercially available hardware
- Modular assemblies

This approach aligned closely with the project's humanitarian and accessibility goals.

Testing Methodology

The wheelchair underwent extensive terrain testing.

The team evaluated performance across multiple terrain conditions including:

- Uphill terrain
- Dirt
- Potholes

Testing involved multiple user groups including:

- Children
- Adult males
- Adult females

This broad testing approach helped validate usability across varying user demographics.

Test Results

The final prototype demonstrated significant performance improvements over standard wheelchairs.

Uphill Performance

- Adventure System: 54.9 seconds
 - Standard Wheelchair: 106.4 seconds
 - Improvement: 49%
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Dirt Terrain

- Adventure System: 16.6 seconds
- Standard Wheelchair: 33.1 seconds
- Improvement: 50%

Pothole Terrain

- Adventure System: 21.7 seconds
- Standard Wheelchair: 24.9 seconds
- Improvement: 12%

Performance Analysis

The strongest performance gains occurred during:

- Hill climbing
- Dirt terrain traversal

This validates the effectiveness of:

- Lever propulsion
- Larger wheel systems
- Terrain-focused geometry

The pothole improvement was smaller but still meaningful, demonstrating improved obstacle handling.

Human-Centered Engineering

One of the most important aspects of the project was designing around real users rather than purely theoretical engineering targets.

The team considered:

- User fatigue
- Accessibility
- Safety
- Repairability
- Ease of use
- Transportation
- Storage

This human-centered approach strongly shaped the final design.

Challenges Encountered

Like most engineering projects, the ELEOS Adventure project encountered multiple design and manufacturing challenges.

Balancing Cost and Performance

One of the largest challenges was improving terrain capability without dramatically increasing manufacturing cost.

Higher-performance systems often require:

- Custom parts
- More complex assemblies
- Specialized manufacturing

The team continuously balanced:

- Simplicity
 - Manufacturability
 - Cost
 - Performance
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Brake Integration

Integrating disc brakes into the wheelchair architecture required:

- Structural mounting considerations
- Wheel compatibility solutions
- User accessibility considerations

The braking system needed to remain:

- Reliable
 - Intuitive
 - Low-maintenance
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Lever System Design

The lever system needed to:

- Transfer force efficiently
- Remain removable

- Prevent excessive complexity
- Avoid interference with wheelchair operation

Designing a robust quick-detach mechanism became an important engineering challenge.

Future Development Opportunities

The team identified several potential future improvements.

Improved Bike Headset

Potential benefits include:

- Lower cost
 - Easier sourcing
 - Simplified assembly
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Better Disc Brakes

Future upgrades could improve:

- Stopping distance
 - User control
 - Safety
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Reinforced Seat Hinge

Potential improvements include:

- Increased durability
 - Longer lifespan
 - Greater structural strength
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Motor Integration

The project explored the possibility of integrating electric motors.

Potential benefits include:

- Higher speed

- Improved hill climbing
- Reduced user fatigue
- Greater accessibility

However, the team noted that motor integration would significantly expand project scope.

Why This Project Matters

The ELEOS Adventure project represents far more than a student engineering prototype.

It demonstrates how engineering can:

- Improve independence
- Increase accessibility
- Restore mobility
- Reduce cost barriers
- Expand outdoor access
- Create practical humanitarian solutions

For many users, mobility directly impacts:

- Education
- Employment
- Healthcare access
- Community participation
- Quality of life

Affordable mobility systems can dramatically change daily life.

Broader Engineering Impact

This project also highlights several broader engineering principles:

Engineering Is About Tradeoffs

Every decision involved balancing:

- Cost
 - Performance
 - Simplicity
 - Manufacturability
 - User experience
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Accessibility Engineering Matters

Engineering projects often focus on performance or innovation alone.

The ELEOS Adventure project demonstrates the importance of designing systems that:

- Real people can afford
 - Real communities can maintain
 - Real environments can support
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Human-Centered Design Leads to Better Engineering

By focusing on real-world users and real-world terrain, the project produced a solution that is practical rather than purely theoretical.

Conclusion

The ELEOS Adventure project successfully developed a low-cost, off-road capable wheelchair platform designed around accessibility, manufacturability, and terrain performance.

Through iterative prototyping, terrain testing, and user-focused engineering, the team created a mobility system capable of significantly improving outdoor accessibility while remaining practical to reproduce and maintain.

The project demonstrates the power of engineering when combined with humanitarian goals and real-world problem solving.

Project Team

ELEOS Adventure Team

- Wyatt Bertis
- Isaiah Jones
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- Caylem Spengler
- Owen Tonseth

Faculty Advisors

- Annika Johnson
- Nathan Slegers

Client

- ELEOS Solutions
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Supporting Documentation

This project included:

- Full assembly documentation
- Prototype development
- Terrain testing procedures
- Design analysis
- Manufacturing planning
- Future development recommendations

The project serves as both a functional engineering prototype and a foundation for future mobility-focused humanitarian engineering development.